

Kehan Guo

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RESEARCH SUMMARY

Ph.D. candidate working on RL post-training, reward and verifier design, and evaluation for reliable LLM agents. Research experience at Apple, Amazon AWS AI, and ByteDance, spanning large-scale post-training, tool-using agents, and scientific reasoning. First/co-first author on two NeurIPS papers, including a Spotlight.

RESEARCH EXPERIENCE

Apple

Jun 2026 – present

Research Intern · Cupertino, CA

- Developing reinforcement-learning post-training methods for large language models, with a focus on reward design and verifiable evaluation.

ByteDance

Apr 2026 – Jun 2026

Research Scientist Intern · San Jose, CA

- Built a tool-using LLM agent for complex information-seeking and risk-investigation tasks, orchestrating domain-specific tools over heterogeneous data sources.
- Developed a feedback-driven skill-evolution pipeline using offline evaluations and real user-query patterns to improve task success.

Amazon AWS AI

May 2025 – Aug 2025

Applied Scientist Intern · New York, NY

- Developed rank-based reward shaping for group-relative policy optimization (GRPO), converting sparse roll-out outcomes into denser within-group feedback for long-horizon reasoning.
- Combined rule-based verification with calibrated model critiques to improve training stability on ambiguous, multi-step tasks.
- Evaluated the approach across mathematical, coding, and logical-reasoning benchmarks.

SELECTED RESEARCH

University of Notre Dame (MINE Lab)

Aug 2022 – Present

Ph.D. Research · Advisor: [Prof. Xiangliang Zhang](#)

- Developed **MolPuzzle**, a multimodal benchmark for molecular structure elucidation; evaluated language and multimodal models against human experts, revealing a large exact-match gap (1.4% best model vs. 66.7% human). NeurIPS 2024 Spotlight.
- Developed **Reward Transport**, using property-aligned noise-data coupling to steer flow-matching models without an inference-time oracle, reward model, gradient, or additional forward pass.
- Contributed to **TRACE**, which compiles natural-language user corrections into runtime checks for coding agents, cutting out-of-distribution preference violations on ClawArena from 100% to 2%.
- Developed benchmarks and agent systems for verifiable scientific reasoning in chemistry, including ChemLLM-Bench (co-first author) and ChemElucid-Live, a live evaluation of tool-using scientific agents.

EDUCATION

University of Notre Dame

Aug 2022 – May 2027 (expected)

Ph.D. in Computer Science

Boston University

2020 – 2022

M.S. in Computer Science

SELECTED PUBLICATIONS

* equal contribution. Full list at [Google Scholar](#).

- [1] Reward Transport: Property Control in Flow Matching via Noise-Space Alignment.
K. Guo, Y. Shen, Y. Zhou, Y. Huang, C. Gao, S. Du, X. Zhang. *Under review 2026* [\[Code\]](#)
- [2] Can LLMs Solve Molecule Puzzles? A Multimodal Benchmark for Molecular Structure Elucidation.
K. Guo, B. Nan, Y. Zhou, et al. *NeurIPS 2024* **[Spotlight]**
- [3] Getting Better at Working With You: Compiling User Corrections into Runtime Enforcement for Coding Agents (TRACE).
Y. Zhou, **K. Guo**, H. Zhuang, et al. *arXiv 2026* [\[arXiv\]](#)
- [4] What Can Large Language Models Do in Chemistry? A Comprehensive Benchmark on Eight Tasks.
T. Guo*, **K. Guo***, B. Nan, et al. *NeurIPS 2023*
- [5] AdaReasoner: Adaptive Reasoning Enables More Flexible Thinking.
X. Wang, Y. Huang, Y. Wang, X. Luo, **K. Guo**, Y. Zhou, X. Zhang. *NeurIPS 2025* **[Spotlight]**
- [6] ChemElucid-Live: Live Evaluation of Tool-Using Scientific Agents.
K. Guo, et al. *Under review 2026*

TECHNICAL SKILLS

Training & Inference: Python, PyTorch, JAX, Megatron-LM, DeepSpeed ZeRO, VeRL, Ray, vLLM, Hugging Face

Methods: RL post-training, GRPO, PPO, DPO, reward modeling, verifier design, LLM evaluation

Agent Systems: Tool use, retrieval, runtime verification, evidence-grounded agents

AWARDS & SERVICE

Awards: NeurIPS 2024 Spotlight, *MolPuzzle* (first author) · NeurIPS 2025 Spotlight, *AdaReasoner* (co-author)
· OpenAI Researcher Access Program (2024)

Reviewer: NeurIPS (2024–25), ICLR (2025–26), ICML, AAI, IJCAI, KDD, WWW, ACL Rolling Review (ACL/EMNLP)

Tutorial: From VAEs to Diffusion and LLMs: Modern Generative Models for Molecular Discovery, KDD 2026

Organizer: [Workshop on Reliable Scientific Foundation Models \(RelSciFM\)](#), KDD 2026